



How to troubleshoot LDAP issues in ONTAP

https://kb.netapp.com/on-prem/ontap/Ontap_OS/OS-KBs/How_to_troubleshoot_LDAP_issues_in_Clust...

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Applies to

- ONTAP 9

Description

- Depending on the configuration of the storage system, Data ONTAP needs to be able to lookup hosts, users, groups, or netgroups information to provide proper access to clients.
- You must configure name-services sources for ONTAP to be able to query local or external name services to obtain this information.

Procedure

The following steps help to troubleshoot LDAP accessibility issues:

Note:

- Following KB examples use IP address '10.64.128.192' as the IP address of the LDAP server.
- Some commands below are available only in diagnostic privilege mode (`set diag`).

1. Verify LDAP as the source in nsswitch:

Before starting the LDAP troubleshooting, it is a good measure to check that LDAP is being configured as a source in the nsswitch configuration.

If LDAP is not specified as a source, it will not be used as a name service even if correctly configured.

```
cluster-1::*> vserver services name-service ns-switch show -vserver
nfstest
Source
Vserver Database Order
-----
nfstest hosts files,dns
nfstest group ldap,files
nfstest passwd ldap,files
nfstest netgroup ldap,files
nfstest namemap files
5 entries were displayed.
```

2. Check that the Storage Virtual Machine (SVM) is able to reach the LDAP server via the IP protocol with the "network ping" command.

```
cluster-1::> network ping -node newnode2 -destination 10.64.128.192
-vserver nfstest

no answer from 10.64.128.192
```

Network configuration and/or infrastructure needs to be checked and corrected if the LDAP server is not reachable.

```
cluster-1::> network ping -node newnode2 -destination 10.64.128.192
-vserver nfstest
10.64.128.192 is alive
```

3. **Check that the Storage Virtual Machine is able to login to the LDAP server using "ldap check"** (command available only in ONTAP 9 and subsequent).

The Storage Virtual Machine will attempt to reach the LDAP server and bind to it using the configured security settings (Minimum binding settings, SSL or Seal and Sign).

If the connection is not successful, a helpful message will be displayed with the reason for it.

Example 1, invalid credentials configured on the LDAP client configuration:

```
cluster-1::*> vservice services name-service ldap check -vservice nfstest
Vservice: nfstest
Client Configuration Name: ldaptest
LDAP Status: down
LDAP Status Details: Error: Validate the Ldap configuration procedure
failed
[ 1 ms] Entry for host-address: 10.64.128.192 not found in the
current source: FILES. Ignoring and trying next available
source
[ 2] Source: DNS unavailable. Entry for
host-address:10.64.128.192 not found in any of the
available sources
[ 2] Resolved LDAP servers: 10.64.128.192. Vservice: 8
[ 2] Failed to initiate Kerberos authentication. Trying NTLM.
[ 3] Successfully connected to ip 10.64.128.192, port 389 using
TCP
[ 8] Unable to connect to LDAP (NIS & Name Mapping) service on
(Error: Invalid credentials)
```

```
[ 8] No servers available for LDAP_NIS_AND_NAME_MAPPING,
vservice: 8, domain: .
**[ 8] FAILURE: Unable to make a connection (LDAP (NIS & Name
Mapping:)), result: 6940
```

Example 2, SSL misconfiguration (specifically lack of root certificate):

```
cluster-1::*> vservice services name-service ldap check -vservice nfstest
Vservice: nfstest
Client Configuration Name: ldaptest
```

```
LDAP Status: down
LDAP Status Details: Error: Validate the Ldap configuration procedure
failed
[ 2 ms] Entry for host-address: 10.64.128.192 not found in the
current source: FILES. Ignoring and trying next available
source
[ 3] Source: DNS unavailable. Entry for
host-address:10.64.128.192 not found in any of the
available sources
[ 3] Resolved LDAP servers: 10.64.128.192. Vserver: 8
[ 4] Successfully connected to ip 10.64.128.192, port 389 using
TCP
[ 4] Unable to start TLS: Server is unavailable
[ 4] Additional info: 00000000: LdapErr: DSID-0C090F01,
comment: Error initializing SSL/TLS, data 0, v23f0
[ 4] Unable to connect to LDAP (NIS & Name Mapping) service on
[ 5] No servers available for LDAP_NIS_AND_NAME_MAPPING,
vserver: 8, domain: .
**[ 5] FAILURE: Unable to make a connection (LDAP (NIS & Name
** Mapping:)), result: 6940
```

See the [NFS configuration guide](#) for specific misconfiguration encountered during this step.

Example 3, A successful connection message will signify that at least a connection at the minimum configured bind level can be achieved:

```
cluster-1::> vservice name-service ldap check -vserver nfstest

Vserver: nfstest
Client Configuration Name: ldaptest
LDAP Status: up
LDAP Status Details: Successfully connected to LDAP server
"10.64.128.192".
```

4. Check that we are able to query information from the LDAP server with the 'getXXbyYY' command.

Lack of resolution might correspond to different situations.

It is possible that the schema configured on the svm differs from the schema configured on the LDAP

server. Check if the contents of "Base DN", "User DN", "Group DN", "Netgroup DN" as well as the specific scopes match the schema being used by the LDAP server.

Another common possibility is that, while it was possible for the svm to reach the LDAP server using the minimum bind level, subsequent queries fail due to lack of privileges.

Raising the minimum bind level and correctly configuring credentials either by specifying correct "Bind DN" and relative password or enabling "Bind Using the Vserver's CIFS Credentials" will fix the issue.

The contents of point 5 can help uncover these kinds of situations.

```
cluster-1::*> getxxbyyy getpwbyname -node newnode2 -vserver nfstest
-username leo -show-source true
(vserver services name-service getxxbyyy getpwbyname)
```

```
Error: command failed: Failed to resolve leo. Reason: Entry not found for
"username: leo".
```

Is very important to specify the "show-source" option in the getxxbyyy command to assure that data seen is coming from the expected source.

An unexpected source used for lookup may correspond to a misconfiguration in nsswitch.

```
cluster-1::*> getxxbyyy getpwbyname -node newnode2 -vserver nfstest
-username root -show-source true
(vserver services name-service getxxbyyy getpwbyname)
Source used for lookup: Files
pw_name: root
pw_passwd: *
pw_uid: 0
pw_gid: 1
pw_gecos:
pw_dir:
pw_shell:
```

If successful, getxxbyyy should produce meaningful data.

```
cluster-1::*> getxxbyyy getpwbyname -node newnode2 -vserver nfstest
-username leo -show-source true
(vserver services name-service getxxbyyy getpwbyname)
Source used for lookup: LDAP
pw_name: leo
pw_passwd:
```

```
pw_uid: 1005
pw_gid: 10000
pw_gecos:
pw_dir: /home/leo
pw_shell: /bin/sh
```

5. In certain scenarios, it can be useful to leverage 'diag secd' command.

The sub-commands for `diag secd` are used for troubleshooting the security daemon. These commands will execute LDAP queries based on the current `ldap` client and `nsswitch` configuration.

In case of failure, a short message can help pinpoint the source of the issue.

More verbose messages will be logged in the `secd.log` file, accessible via the SPI interface.

To collect such logs, see KB [How to manually collect logs and copy files from a clustered Data ONTAP storage system](#).

The following example shows an attempt to translate a unix user name to the corresponding user ID. To do so, Secd will need to reach the the LDAP server and query the contents of the "passwd" database for the specified user.

```
cluster-1::*> diag secd authentication translate -node newnode1 -vserver
nfstest -unix-user-name leo
```

```
Vserver: nfstest (internal ID: 8)
```

```
Error: Acquire UNIX credentials procedure failed
[ 61 ms] Entry for host-address: 10.64.128.192 not found in the
current source: FILES. Ignoring and trying next available
source
[ 70] Source: DNS unavailable. Entry for
host-address:10.64.128.192 not found in any of the
available sources
[ 71] Resolved LDAP servers: 10.64.128.192. Vserver: 8
[ 93] Entry for host-name: newnode1 not found in any of the
available sources
[ 105] Failed to initiate Kerberos authentication. Trying NTLM.
[ 111] Successfully connected to ip 10.64.128.192, port 389 using
TCP
[ 221] LDAP search for the "uid, uidNumber, gidNumber,
```

```
unixUserPassword, name, unixHomeDirectory, loginShell"
attribute(s) within base "dc=leodom2,dc=root" (scope: 2)
using filter "(&(objectClass=User)(uid=leo))" failed with
error: Operations error
[ 221] Additional info: 000004DC: LdapErr: DSID-0C090724,
comment:
```

In order to perform this operation a successful

bind must be completed on the connection., data 0, v23f0

```
[ 221] Source: LDAP unavailable. Ignoring and trying next
available source for user-name: leo
```

```
[ 231] Entry for user-name: leo not found in the current source:
FILES. Entry for user-name: leo not found in any of the
available sources
```

```
**[ 231] FAILURE: Unable to retrieve UID for UNIX user leo
```

In this specific example, the issue is related to the configuration of credentials. Check point 4 on how to proceed in these situations.

6. As a last resort, a packet trace will show the communication between the storage controller and LDAP server.

The following articles describes the procedure to capture a packet trace on the storage controller:

- [How to capture packet traces \(tcpdump\) on ONTAP 9.2+ systems](#)
- [How to capture packet traces \(PKTT\) on Data ONTAP 8 Cluster-Mode systems](#)

Additional Information

N/A